

IAL Biology Unit 4 (WBI14/01) — Complete A* Study Note

Energy, Environment, Microbiology and Immunity

Built entirely from analysis of 21 past papers and their mark schemes, June 2019 → June 2026.

Everything below is traceable to what examiners actually wrote. Where a mark scheme states an **ACCEPT**, **IGNORE** or **DO NOT ACCEPT** ruling, it is quoted verbatim — those rulings are where marks are won and lost. This note is designed to be the only thing you need.

0. THE PAPER, AND WHAT THE MARK SCHEMES REWARD

Code	WBI14/01
Time	1 hour 45 minutes
Marks	90
Structure	8–9 questions , no sections; multiple-choice, short-open, open-response, calculation, extended writing
Asterisked (*) questions	1–3 per paper, marked by levels , not points
Maths	Minimum 9 marks at Level 2 or above
Synoptic	Questions may draw on Units 1 and 2
Equipment	Scientific calculator, ruler, HB pencil

Pace: 90 marks in 105 minutes $\approx$ **1.17 min per mark**. Use the mark allocation in brackets as your timer.

Topic weighting, counted across the 21 papers

Topic	Marks/paper	Appears in
Photosynthesis (LDR, Calvin cycle, chloroplast, pigments)	12–20	Every paper
Immunity and the immune response	10–18	Every paper
Microbiology: culture, growth curves, viruses	10–18	Every paper
Climate change: evidence, causes, effects	8–16	Every paper
Energy flow: GPP, NPP, trophic efficiency	6–14	Every paper
Antibiotics and resistance	6–14	~90%
Evolution, natural selection, speciation	5–12	~90%
Ecology: niche, succession, sampling	5–12	~85%
Forensics: time of death	4–10	~75%
Decomposition and carbon recycling	3–8	~70%
PCR, electrophoresis, DNA profiling	3–10	~60%

The four things the mark schemes reward that students miss

1. **Exact terminology.** Biology mark schemes list rejections. *“DO NOT ACCEPT immune for resistant.” “DO NOT ACCEPT to T killer cells.” “IGNORE grown in a Petri dish.”* Getting the concept right is not enough; you must use the accepted word.
2. **Linked chains, not lists.** One mark scheme states outright: *“NB answer must be linked with a time delay / many-staged process / slow process for max 4 marks to be awarded.”* Unlinked facts are capped.
3. **Data quoted with figures.** “Describe the data” marks require **numbers from the graph or table**, with units.
4. **Explanations that go one step further than the observation.** In the levels-based questions, mark schemes tag points as **(D) description** or **(E) explanation**, and Level 3 requires the explanations.

PART A — PHOTOSYNTHESIS AND ENERGY

1. Where the reactions happen

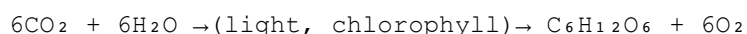
Asked as a multiple-choice question in most papers (e.g. Jan 2021 Q1a, Oct 2020 Q1a, Oct 2023 Q1):

Stage	Location
Light-dependent reactions	Thylakoid membranes
Light-independent reactions (Calvin cycle)	Stroma

The mark scheme's own explanation of the distractors: *"A is incorrect because the light-dependent reactions take place in the thylakoid membranes... D is incorrect because the light-independent reactions take place in the stroma."*

Overall reaction, in the words the specification and mark schemes use:

Light energy is used to **split apart the strong bonds in water molecules**; the **hydrogen is stored in a fuel (glucose)** by combining it with **carbon dioxide**; **oxygen is released** into the atmosphere.



⚠ **The atom-origin question (Jan 2021 Q5b(ii)) — a 2-marker with a stated rejection:**

Carbon and oxygen in glucose come from carbon dioxide. Hydrogen comes from water. Mark scheme: *"ACCEPT CO₂ ... ACCEPT H₂O ... DO NOT ACCEPT O from water."*

The oxygen released as gas comes from water; the oxygen **in the glucose** comes from carbon dioxide. Students routinely get this backwards.

2. Chloroplast structure — the drawing and labelling question

Asked directly (Jan 2021 Q1b, Oct 2021 Q1a, Oct 2023 Q1, May 2025 Q6). The Jan 2021 mark scheme awards marks for drawing and labelling structures, with *"IGNORE lipid droplets, stroma, thylakoid membranes"* — meaning you must name the **less obvious** structures:

Structure	Role — as examiners phrase it
Double membrane (envelope)	Controls entry and exit of substances

Thylakoid membranes, stacked into grana	Hold the photosystems, pigments, electron carriers and ATP synthase ; large surface area for light absorption
Thylakoid space	Small volume, so the proton gradient builds quickly
Stroma	Site of the light-independent reactions ; contains RUBISCO and Calvin-cycle enzymes
DNA loop — “ <i>ACCEPT plasmid / plasmid-like DNA</i> ”	Codes for some of the chloroplast's own proteins
70S ribosomes	Synthesise those proteins
Starch grain	Stores the product of photosynthesis

Magnification (Oct 2021 Q1a(iii)):

$$\text{magnification} = \text{image size} \div \text{actual size}$$

Measure in **mm**, convert to **µm (× 1000)**, then divide. Show the conversion.

Chloroplast vs mitochondrion comparison (June 2021 Q2a): both have a **double membrane, circular DNA, 70S ribosomes** and **internal membranes with a large surface area**. Differences: chloroplast has **grana/thylakoids and pigments**, mitochondrion has **cristae**; chloroplast has **stroma**, mitochondrion has **matrix**; chloroplast contains **starch grains**.

3. The light-dependent reactions

3.1 The chain examiners want

1. **Light energy is absorbed by chlorophyll** in the photosystems.
2. **Electrons are excited** and released from the chlorophyll/photosystem.
3. Electrons pass along an **electron transport chain**, releasing energy.
4. Energy is used to **pump protons into the thylakoid space**, creating a **proton gradient**.
5. Protons diffuse back through **ATP synthase — chemiosmosis** — producing **ATP**.

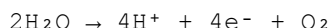
6. Electrons and protons are accepted by **NADP**, forming **reduced NADP**.

⚠ The mark scheme for the pigment question (Mar 2020 Q7a(ii), Jan 2022 Q6b) is only 2 marks and both points are fixed:

• "to absorb light so that the electrons will be **excited / released from photosystems**" •
"so that **reduced NADP and ATP** can be synthesised" — "ACCEPT NADPH /
photophosphorylation"

Learn those two clauses verbatim. That exact pair has appeared at least three times.

3.2 Photolysis — its three roles



From the June 2019 Q7a mark scheme, the creditable points are:

• **water is split by light** — "reference to photolysis" • producing **H⁺ and electrons** —
"ACCEPT protons" • to **replace electrons lost by chlorophyll** • to **provide protons for**
the formation of reduced NADP • **oxygen** is released

And the follow-through the mark scheme wants: "reduced NADP used in the conversion of GP to GALP." The chain from water to sugar must be completed.

3.3 Cyclic versus non-cyclic photophosphorylation

Tested in Jan 2021 Q5b(i) and June 2021 Q2b.

	Non-cyclic	Cyclic
Photosystems	PSII and PSI	PSI only
Electron source	water (photolysis)	recycled to the same chlorophyll
Products	ATP, reduced NADP, oxygen	ATP only
Photolysis	Yes	No

The Jan 2021 mark scheme rejects distractors with: "ATP is not available to light-dependent reactions in cyclic photophosphorylation" and "oxidised NADP is not produced."

Named product of non-cyclic only (June 2021 Q2b(ii)): reduced NADP (or oxygen).

4. The Calvin cycle

- | | |
|--|-----------------|
| 1. $\text{CO}_2 + \text{RuBP (5C)} \rightarrow [\text{RUBISCO}] \rightarrow 2 \times \text{GP (3C)}$ | CARBON FIXATION |
| 2. $\text{GP} + \text{reduced NADP} + \text{ATP} \rightarrow \text{GALP (3C)}$ | REDUCTION |
| 3. 5 of every 6 GALP + ATP $\rightarrow$ regenerate RuBP | REGENERATION |
| 1 of every 6 GALP leaves the cycle to make products | |

Per hexose: **6 turns · 6 CO₂ · 18 ATP · 12 reduced NADP.**

4.1 The “why does the rate change?” chain

June 2019 Q7c(i) — plants in light versus dark. The mark scheme awards 3 of these 5 linked points:

1. more **light energy for the light-dependent reactions** — *“ACCEPT light is not a rate-limiting factor”*
2. more **photolysis**
3. more **electrons released/excited from chlorophyll/photosystems**
4. more **ATP and reduced NADP** produced
5. so the **light-independent reactions / Calvin cycle / carbon fixation / GP production is faster** — *“RuBP regenerated faster to bind more CO₂”*

△ **The reverse case (June 2019 Q7c(ii)) — why net CO₂ uptake is negative:**

- **respiration is greater than photosynthesis** — *“ACCEPT more carbon dioxide being released by respiration than being used in photosynthesis”; “DO NOT ACCEPT no photosynthesis”*
- **RuBP regenerated too slowly to bind as much CO₂**
- **because the light intensity is very low**

That third clause — *RuBP regenerated too slowly* — is the discriminating mark and appears in several papers.

4.2 What GALP becomes — the mark scheme’s own route

Mar 2020 Q7b(i), “how GALP is converted into protein”, 3 of 4 points:

- **GALP converted into hexose/glucose/sugar**
- **amino acids made from glucose and nitrates** — *“IGNORE other nitrogenous compounds”*
- **amino acids joined by peptide bonds / condensation reactions**
- **glucose respired to produce ATP for protein synthesis**

Other routes to know:

Product	Route
Starch, cellulose, sucrose	glucose joined by condensation , forming glycosidic bonds
Lipids	GALP → glycerol and fatty acids ; condensation forming ester bonds
Nucleic acids	sugar + phosphate + nitrogenous bases

⚠ **The recurring trap:** “*explain why amino acids cannot be produced from glucose alone*” — because glucose contains only **carbon, hydrogen and oxygen**, and amino acids also contain **nitrogen**, obtained as **nitrate ions from the soil**.

Starch vs cellulose (June 2019 Q5b) — two similarities and two differences:

- *Similarities:* both **polysaccharides**; both **polymers of glucose**; both made by **condensation**; both joined by **glycosidic bonds**.
- *Differences:* starch is **α -glucose**, cellulose **β -glucose**; starch is **branched/coiled**, cellulose is **straight chains with alternate molecules inverted**; cellulose forms **microfibrils held by hydrogen bonds**.
- *Why each is where it is:* starch is **compact and insoluble**, so it stores energy **without affecting water potential**; cellulose is **strong** because of hydrogen bonding between chains, so it supports the **cell wall**.

5. Pigments, spectra and chromatography

🔗 **Absorption spectrum:** a graph of **how much light is absorbed** at each wavelength. 🔗

Action spectrum: a graph of the **rate of photosynthesis** at each wavelength.

The two have **similar shapes** — peaks in the **blue (~450 nm)** and **red (~660 nm)**, a trough in the **green** — which is **evidence that the pigments absorbing the light are the ones driving photosynthesis**. Leaves look green because green light is **reflected/transmitted, not absorbed**.

Sketching an action spectrum onto an absorption spectrum (Mar 2022 Q2a(i)) — the mark scheme:

- “a line that roughly follows the contours of the absorption spectrum lines and **does not drop to zero**”
- “the **left-hand peak (at about 460)** is **higher** than the right-hand peak(s)”

Why plants have several pigments: different pigments absorb **different wavelengths**, so a **wider range of the spectrum** is used, increasing the rate.

Chromatography and Rf:

$Rf = \text{distance moved by the pigment} \div \text{distance moved by the solvent front}$

Measure from the **origin to the centre of the spot**.

Pigment	Colour	Rf
Carotene	orange	highest
Xanthophyll	yellow	high
Chlorophyll a	yellow-green	lower
Chlorophyll b	blue-green	lowest

Why carotene travels furthest: it is **most soluble in the solvent** and **least strongly attracted to the stationary phase**.

Method marks: grind with **sand and a small volume of propanone**; origin line in **pencil**; apply the spot **repeatedly, drying between**; keep the **origin above the solvent level**; use a **lid** to keep the atmosphere saturated.

Interpreting extraction data (Jan 2022 Q6c *): the mark scheme wants **direct comparisons** — “*propanone extracts more chlorophyll a from species P ... DMSO extracts more...*” — i.e. compare **solvent by solvent and species by species**, with figures.

6. Investigating the rate of photosynthesis

Use an aquatic plant (*Elodea*, *Cabomba*) in **sodium hydrogencarbonate solution** as the CO₂ source.

Two measurement methods:

1. **Count bubbles** per minute — simple, but bubbles vary in size.
2. **Photosynthometer** — collect the gas and measure the **length of the gas column**; more reliable because it measures actual volume.

Variable	How to change it	Critical control
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Light intensity	move a lamp; plot against $1/d^2$	heat shield between lamp and plant
Wavelength	coloured filters or LEDs	keep light intensity the same (filters absorb light)
Temperature	water bath	light intensity, CO ₂
CO₂	hydrogencarbonate concentration	light, temperature

Marks come from: the **heat shield** (so only light intensity changes, not temperature); **5 minutes to equilibrate** at each new condition; cutting the stem **freshly under water**; **repeats and a mean**; and noting that the gas collected is **not pure oxygen** and some **dissolves in the water**, so the volume underestimates the rate.

The classic graph: rate rises with light intensity then **plateaus**, at which point **another factor is limiting** — CO₂ or temperature.

7. GPP, NPP and energy transfer

7.1 The definitions and the units

Term	Meaning
GPP	The total energy/biomass fixed by photosynthesis per unit area per unit time
R	Energy used by the plants in respiration
NPP	$GPP - R$ — what is left for the next trophic level

Units: $\text{kJ m}^{-2} \text{yr}^{-1}$. The June 2019 Q5a(ii) mark scheme rejects every other option with “*GPP is the energy per unit area in a given time*” — the units **must contain both an area and a time**.

⚠ In a table, **GPP is always the largest**, and $GPP = NPP + R$. The June 2019 mark scheme rejects two distractors purely with “*GPP is the highest*”, and a third with “*respiration is not energy in a trophic level.*”

7.2 The efficiency calculation

$$\text{efficiency (\%)} = (\text{energy in the higher trophic level} \div \text{energy in the lower}) \times 100$$

Worked, from the June 2019 Q5a(iii) mark scheme itself: $(6 \times 100) \div 67 = 8.96 \%$.

Note the scheme rejects a distractor solely because “8.955522388059701 rounds up to 8.96” — **round correctly**.

Other worked examples: June 2021 Q3b (lions, giraffes, acacia: $4656 \div 44\,988 \times 100$); Oct 2019 Q6a (wolf 65 kg, moose 620 kg, ratio **1 : 9.5** — “ACCEPT 1 : 9.54, 10 : 95, 0.1 : 1; IGNORE correct fractions”).

7.3 Why so little energy is transferred

June 2019 Q5a(iv) — the mark scheme’s answer is simply “energy is transferred to the environment”, and it rejects the alternatives with “you cannot make energy”, “not all organisms increase in size as you go up trophic levels” and “there are not always fewer organisms at higher trophic levels.”

The full lists for longer questions:

Sunlight to producers: light **misses the leaves** or is **reflected/transmitted**; some is the **wrong wavelength**; light, CO₂ or temperature may be **limiting**; energy is **lost in respiration**.

Between consumers: not all is **eaten** (bones, roots, bark); not all eaten is **digested/absorbed** — some is **egested in faeces**; energy is **lost as heat in respiration**, especially in **endotherms**; energy is **lost in excretion (urea)**.

Consequences: food chains are **short** (rarely more than 4–5 links), and it is **more energy-efficient for humans to eat plants than meat**.

PART B — ECOLOGY AND THE ENVIRONMENT

8. The ecological definitions — mark-scheme wording

These are 1-mark questions asked in nearly every paper. The rulings matter.

Term	The accepted definition
Habitat	The place where an organism lives

Population	All the organisms of one species in a habitat at one time
Community	All the populations of different species living and interacting in a habitat
Ecosystem	<i>"organisms interacting with each other AND with the (physical/chemical) environment"</i> — <i>"ACCEPT species/plants and animals/biotic factors", "ACCEPT abiotic factors/non-living environment", "IGNORE habitat"</i>
Niche	<i>"the role that an organism plays (in its habitat)"</i>

⚠ **Ecosystem needs BOTH halves** — organisms *and* the abiotic environment. "Habitat" is explicitly ignored.

⚠ **The niche question needs a stated reason.** In Mar 2020 Q6a, on two barnacle species, the mark scheme awards: *"niche is the role that an organism plays"* + *"the two types of barnacles are found on different parts of the rock"* + **"credit a stated reason"** — e.g. *"food sources / temperature tolerances / desiccation rates."* A definition alone will not score 3.

Biodiversity (Jan 2022 Q6a) — two marks, two components:

- **number of different species** — *"ACCEPT species richness"* • **genetic diversity within a species** — *"ACCEPT variation in genotypes/alleles"*

Abundance vs distribution (Oct 2019 Q8a): **abundance** = the **number of individuals**; **distribution** = **where** they are found / how they are spread.

Biotic factors: predation, competition, disease, food availability, parasitism. **Abiotic factors:** temperature, light, water, pH, salinity, oxygen, soil type, wind, mineral ions.

Competitive exclusion: no two species can occupy exactly the same niche — one would **outcompete** the other, which would be **excluded** or evolve a **slightly different niche**.

9. Ecological sampling

Quadrats — for abundance:

- **Random sampling** to avoid bias: lay two tapes as **axes**, generate **random coordinates**, place the quadrat there.
- Record **density, frequency** (% of quadrats containing the species) or **percentage cover**.
- Take **enough samples** — plot a **species–area curve** and use the plateau.

estimated population = mean per quadrat × (total area ÷ quadrat area)

Transects — for distribution along a gradient (sand dune, rocky shore, woodland edge). Use a **line** or **belt** transect. Use them where distribution is **not uniform**, because random sampling would hide the pattern.

Abiotic measurements alongside: light meter, thermometer/probe, pH meter, soil moisture (or **mass before and after drying to constant mass**), anemometer, oxygen probe. Take readings at the **same time of day**.

“Comment on the reliability” — the mark schemes’ recurring points:

Was the **sample size** large enough? Were there **repeats**? Was sampling **random**? Were **variables controlled**? Was there a **control group**? Are differences large enough to be significant? Was only **one habitat / one time / one species** studied? **Correlation does not prove causation**.

10. Succession

📌 **Succession:** the change in a community over time, from colonisation of bare ground to a stable **climax community**.

The chain:

1. **Pioneer species** colonise bare rock/sand; they tolerate harsh abiotic conditions.
2. They **die and are decomposed**, adding **humus** and forming **thin soil**.
3. The soil **retains more water and nutrients**, so conditions become **less hostile**.
4. **New species colonise and outcompete** the pioneers.
5. **Biodiversity increases** at each stage (sere).
6. A **stable climax community** is reached.

Primary succession starts with **no soil**; **secondary succession** starts where soil remains (after fire or clearance) and is **faster** because soil, nutrients and a **seed bank** are already present.

The sand-dune * question (June 2021 Q7a(iii)) — how the levels work

The mark scheme tags every point **(D)** for description or **(E)** for explanation, and states the levels explicitly:

"Level 1: 1 mark = 1 description; 2 marks = 3 descriptions OR 1 description + an explanation." "Level 3: 5 marks = 4 descriptions and all 4 explained; 6 marks = 5 descriptions and all five explained."

This tells you exactly how to write it: pair every observation with its explanation.

Description (D)	Explanation (E)
Soil depth increases with distance from the sea	These are older dunes , so there has been more time for humus to build up
Organic material increases	More plants have grown and died and been decomposed
pH falls with distance from the sea	More humus is added; rain washes out minerals (leaching of shell fragments)
Percentage of bare rock/sand decreases	Pioneer species break it down; more soil covers it
Number of plant species increases	The soil is more fertile and salinity is decreasing ; organic material holds more moisture
Types of plant change	Soil becomes more favourable for more species , and those plants improve conditions further

Forest fire and secondary succession (Jan 2023 Q9d *): the mark scheme lists "*(secondary) succession begins → increasing biodiversity → as biodiversity of plants increases so does the biodiversity of animals → eventually climax community re-established.*"

11. Evidence for climate change

Four kinds, each with its limitation.

Evidence	Mechanism	Limitation
CO₂ records	Direct measurement, and air bubbles trapped in ice cores , dated by annual layers	Only from polar regions; dating uncertainty at depth
Temperature	Direct thermometers (~150 yr); before that, ¹⁸O:¹⁶O isotope ratios in ice cores — more ¹⁸ O	Short record; uneven station

records	= warmer	distribution; urban heat islands
Pollen in peat bogs	Peat is acidic and anaerobic , so decomposition is very slow and pollen is preserved ; grains have species-specific shapes ; deeper = older; carbon dating gives the age; knowing each species' climate requirements gives the past climate	Pollen may be blown in from elsewhere; not all plants produce durable pollen; tolerances may have changed
Dendrochronology	One ring per year; wider rings = warmer/wetter growing season; overlapping samples extend the record	Ring width also depends on nutrients, disease, competition, shading — not temperature alone

Identifying the pollen species (Oct 2019 Q2b(ii)) — the mark scheme's three points:

- **isolate DNA** from the pollen • use **gel electrophoresis** — *"ACCEPT detail that includes applying a current to a gel"* • **match the pollen DNA profile/bands to those of known plants** — *"IGNORE fragments"*

△ **Correlation and causation.** Mark schemes award a mark for stating that these data show a **correlation**, and that establishing **causation** needs a **mechanism** and the exclusion of other variables.

12. Anthropogenic climate change and the greenhouse effect

📌 **The verbatim mark-scheme definition (Jan 2021 Q4a(ii)) — two marks, two clauses:**

- *"(anthropogenic) **caused by the effects of humans**" — "IGNORE named activities"* • *"(climate change) **changes to (mean) temperature / rainfall**" — "ACCEPT long-term (mean) change in weather patterns"; "IGNORE weather unqualified"*

Listing "burning fossil fuels" earns nothing. You must state **caused by humans** and **change in mean temperature or rainfall**.

The greenhouse effect — the four-step chain:

1. **Short-wavelength radiation** from the Sun **passes through the atmosphere** and is absorbed by the Earth's surface.

2. The surface **re-radiates energy at longer wavelengths (infrared)**.
3. **Greenhouse gases absorb this infrared and re-radiate some back towards the Earth.**
4. Less energy escapes, so the **mean surface temperature rises**.

Greenhouse gases: carbon dioxide, **methane**, water vapour, nitrous oxide, CFCs. **Δ The standard MCQ rejection:** “D is incorrect because **oxygen is not a greenhouse gas**.” Nor is nitrogen.

The methane chain (Oct 2019 Q8b–c) — the mark scheme’s exact points:

• microorganisms **secrete enzymes to break down organic matter** — “*ACCEPT named enzyme breaking down named organic molecule*” • **glucose/simple sugars used for respiration** by microorganisms • **anaerobic conditions/respiration result in methane being produced** — “*IGNORE carbon dioxide*” • **methane is a greenhouse gas** — “*ACCEPT methane traps infra-red radiation/heat energy*” • resulting in an **increase in temperature of the atmosphere/surface/water** — “*IGNORE global temperatures*”

That last rejection is worth noting: you must specify **what** is warming.

The knock-on effects the same scheme credits: “*increase in abundance due to increased growth/development/NPP/biomass/GALP in warmer conditions*”, “*as increased enzyme activity/RUBISCO/photosynthesis*”, and “*increase in distribution as colder areas become warmer*” — “*IGNORE distribution*” under the abundance point, so keep the two separate.

13. The carbon cycle and reducing atmospheric CO₂

ATMOSPHERIC CO₂

↓ photosynthesis	↑ respiration (plants, animals, decomposers)
PLANTS → feeding → ANIMALS	↑ combustion of fossil fuels and biomass
↓ death, excretion	↑ exchange at the ocean surface
DEAD ORGANIC MATTER → decomposition → fossilisation → FOSSIL FUELS	

“Explain one way of reducing the effects” (June 2021 Q7c(ii)) — the mark scheme gives three alternatives, each worth 2 marks and each needing a mechanism:

• **reduce deforestation** — “*because this will reduce the carbon dioxide released into the atmosphere*” • **reforestation** — “*because more plants will absorb more carbon dioxide for photosynthesis*” • **reduce the number of cattle farmed** — “*ACCEPT reduce the extent of rice farming*” — “*as this will reduce the methane being released*”

Naming the method alone scores 1; the **mechanism** scores the second mark.

Other methods: carbon capture and storage; **biofuels** (the CO₂ released was **recently removed by photosynthesis**, so approximately carbon neutral); preserving **peat bogs and wetlands**, which are large carbon stores.

14. Extrapolation, models and their limits

“Describe how this graph could be used to estimate the value in 2020” (June 2019 Q8a(ii)):

Extend the line of best fit beyond the data (extrapolate) and **read off** the value — *“ACCEPT by extrapolation for 1 mark.”*

“Explain why the prediction may be unreliable” (June 2019 Q8a(iii)) — 3 of these 5:

• the **data fluctuates** — *“ACCEPT 2020 might be on a peak or trough”* • a factor that **decreases** global warming could change — *“ACCEPT efforts are being made to decrease global warming”* • a factor that **increases** global warming could change — *“e.g. volcanic eruptions, forest fires, albedo effect”* • the **trend could change** — *“the prediction assumes that the trend will not change”* • **data collected long ago may be less accurate** due to the technology available

Drawing a correlation graph (Jan 2021 Q9a(iii)): for a negative correlation the mark scheme wants *“a straight line sloping down from top left to bottom right”*, with axes labelled (moose number against temperature), and notes *“ALLOW a double y-axis graph correctly labelled + line for three marks.”*

15. Effects of climate change on organisms

Structure any answer around the three categories.

(a) Distribution — species move **towards the poles or to higher altitudes**; those already at the poles or on mountain tops have **nowhere to go**; migration may be **too slow** or **blocked by human development**.

(b) Development — most organisms are **ectothermic**, so higher temperature → **higher rate of enzyme-controlled reactions** → **faster development and growth**, up to the optimum, beyond which enzymes **denature**.

(c) Life cycles — earlier flowering, egg-laying, insect emergence and migration. The danger is **desynchronisation**: if an insect emerges before its food plant, or chicks hatch after the

caterpillar peak, populations fall.

The polar bear question (June 2019 Q8b) is the model application: less sea ice → fewer hunting platforms → bears eat **lower-energy foods** such as berries → an enormous number of berries is needed for the same energy → **energy intake is insufficient** → body mass, reproduction and survival fall. Note the mark scheme's phrasing "*more energy for insulation*" and "*fat used for insulation*."

Temperature and reproduction (Jan 2021 Q4c): read the graph, identify the **optimum**, and explain differences between sexes in terms of **enzyme denaturation** at high temperature.

16. Temperature, enzymes and Q_{10}

The enzyme explanation:

Below the optimum: higher temperature gives molecules more **kinetic energy**, so they **move faster** and there are **more frequent successful collisions**, so **more enzyme–substrate complexes** form and the rate rises. **Above the optimum:** increased vibration **breaks hydrogen and ionic bonds** in the tertiary structure, so the **active site changes shape**, the substrate **no longer fits**, and the enzyme is **denatured**.

$$Q_{10} = \text{rate at } (T + 10\text{ }^{\circ}\text{C}) \div \text{rate at } T\text{ }^{\circ}\text{C}$$

Worked from the Mar 2024 Q2b(ii) mark scheme: "*values for R20 and R10 read from the graph (6 and 21.5/22)*" then " *Q_{10} given as 4*", with two explicit rulings: "**DO NOT ACCEPT with units**" and "*Bald answer of 4 = 2 marks*."

⚠ **Q_{10} has no units**, and it is only meaningful **below the optimum**.

The C_3 / C_4 comparison in the same paper wants: "*rate increase in C_3 is less steep*" and " *C_3 plants probably photosynthesise at lower temperatures*."

Investigating temperature and development (brine shrimp or seedlings): counted eggs/seeds in identical volumes of solution at a range of temperatures in **water baths**; fixed time; count **number hatched** or measure **shoot length/dry mass**. Control **salinity, light, pH, oxygen, batch and number of organisms**. **Ethics:** use **invertebrates or eggs**, avoid temperatures causing suffering, use the **minimum number**, dispose humanely, never release non-natives.

17. Evolution by natural selection

📌 Evolution = a change in allele frequency in a population.

The chain — from the Jan 2021 Q6a(i) mark scheme, worth 3 marks:

• *"mutation (in the DNA) causing allele for resistance to the new chemical" — "ACCEPT forming a (new) gene for resistance"* • *"(new) chemical acts as a selection pressure"* • *"resistant flies survive and pass this gene/allele onto their offspring"*

⚠ The same mark scheme carries TWO explicit rejections, both of which cost students marks every session:

"DO NOT ACCEPT immune for resistant." *"DO NOT ACCEPT selection pressure causes mutation."*

Mutation is **random** and occurs **before** the selection pressure acts. Never write that the pressure caused the mutation, and never write "immune" when you mean "resistant".

"Why does resistance evolve so quickly?" (Jan 2021 Q6a(ii)) — any one of:

short life cycle • high mutation rate • many offspring produced • the chemical is a strong selection pressure

Antibiotic resistance and prescription data (Oct 2020 Q7b) — the mark scheme's four points:

• there is a **correlation** between the number of prescriptions and the percentage of resistant *E. coli* — *"ACCEPT pattern/trend"*, *"IGNORE directly proportional"* • the antibiotic acts as a **selection pressure** • therefore the **resistant bacteria reproduce and the non-resistant die** — *"ACCEPT the resistant bacteria are more likely to reproduce"* • the percentage of resistant bacteria **falls when prescriptions fall**, because non-resistant bacteria are **not destroyed** • *"credit a comment about competition between resistant and non-resistant bacteria"*

Bacteria also spread resistance **horizontally, by conjugation via plasmids**.

Genetic diversity and gene flow (Mar 2022 Q1d):

genetic diversity **decreases** • because **gene flow is reduced** — *"ACCEPT fewer allele combinations passed on"* • as a result of **inbreeding** — *"IGNORE refs to mutations increasing diversity."*

18. Speciation

	Allopatric	Sympatric
Isolation	Geographical — a physical barrier	Within the same area — ecological, behavioural, temporal
Mark-scheme example	<i>"lemurs have the same common ancestor isolated from those in Africa"</i> (June 2021 Q4a)	<i>"Sifakas and Indri live together but have a different diet"</i> (same question)

"ACCEPT geographic speciation" for allopatric.

The full * chain (Aug 2022 Q5) — the mark scheme's own sequence:

• organisms moved apart depending on where their food was • **sympatric speciation / organisms not separated by a physical barrier** • **therefore reduced gene flow** between the two groups • **resulting in changes in the allele frequency** • eventually resulting in **reproductive isolation** • **and the formation of two species**

The levels note: *"6 marks = 7 points made which includes at least one level 3 point."*

△ **Read which is being asked.** Papers routinely place *"explain how X evolved feature Y"* (§17's chain) next to *"explain how X became a new species"* (§18's chain). They are different questions carrying different marks.

19. Sustainability, reforestation and controversy

Reforestation as conflict management (June 2023 Q8):

Human needs: **timber, fuel, paper**, jobs, harvest on rotation. *Conservation:* trees **photosynthesise, removing CO₂**; provide **habitats**, increasing **biodiversity**; **roots stabilise soil**, preventing erosion; **reduce flooding**. *Management:* **selective logging** and **coppicing** not clear-felling; **replant** at least as many as are cut; **native, mixed species** not monoculture; leave **wildlife corridors**.

Biofuels: approximately **carbon neutral** because the CO₂ was **recently absorbed by photosynthesis**; renewable. But **land is used for fuel not food**; **habitats cleared**; energy used in **growing, processing and transport** means they are **not truly neutral**.

Why scientific conclusions depend on who reaches them: researchers may be **funded** by interested parties; different groups **select different data** or use **different models**; **political and economic priorities** differ; media reports may be **unreviewed or exaggerated**; **peer review and replication** are the safeguards.

PART C — MICROBIOLOGY, IMMUNITY AND FORENSICS

20. Culturing microorganisms

"Describe one way bacteria can be grown in a laboratory" (June 2019 Q3a) — the mark scheme's four points, 3 required:

• *"on agar / in a broth / in nutrients"* — **"IGNORE grown in a Petri dish"** • *"use of aseptic technique / description of aseptic technique"* — "e.g. sterile equipment, autoclaved media, working by a Bunsen burner" • **"incubated at a stated temperature in the range of 25 °C to 35 °C"** • *"credit another detail of culture conditions"* — "e.g. appropriate pH, appropriate growth factors, appropriate length of time in the range of 24 hours to 1 week, description of ensuring that conditions are not anaerobic"

⚠ **You must name the medium and state an actual temperature.** "In a Petri dish at a suitable temperature" scores zero of those two marks.

Aseptic technique — the full list

1. STERILISE equipment and media by AUTOCLAVING (121 °C, 15 min)
2. DISINFECT the work surface before and after
3. Work CLOSE TO A BUNSEN FLAME — the updraught carries airborne microbes away
4. FLAME the inoculating loop before and after use, and cool it
5. FLAME the neck of bottles when opening and closing
6. Lift the Petri dish lid only SLIGHTLY and at an ANGLE
7. Tape the lid in two or four places but DO NOT SEAL IT COMPLETELY — sealing creates ANAEROBIC conditions in which harmful anaerobic pathogens could grow
8. Incubate INVERTED at no more than 25 °C in schools — reduces the risk of growing human pathogens and stops condensation dropping onto colonies
9. AUTOCLAVE everything before disposal

Why: to prevent **contamination of the culture** (which would compete and invalidate results) and **contamination of the environment and the experimenter**.

21. Measuring microbial growth

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Method	Advantage	Disadvantage
Cell count (haemocytometer)	Quick; absolute number	Counts living and dead ; miscounting
Dilution plating (viable count)	Counts only living cells ; detects low numbers	Assumes one colony = one cell ; takes 24–48 h
Mass (dry mass to constant mass)	Direct measure of biomass	Destructive; slow
Turbidity / optical density	Quick, non-destructive , continuous	Counts living and dead and debris; needs a calibration curve

$$\text{cells per cm}^3 = (\text{number of colonies} \div \text{volume plated}) \times \text{dilution factor}$$

Use a plate with 30–300 colonies: below 30 is **statistically unreliable**; above 300 colonies **merge and are miscounted**.

⚠ **The * question (Mar 2020 Q9e): why one calibration curve cannot serve all microorganisms.**

Different species have **different cell sizes and shapes**, so they **scatter and absorb light differently** — a large yeast cell affects optical density far more than a small bacterium. Species also differ in **pigmentation**. So the **same optical density corresponds to a different number of cells** in each species, and each needs its **own calibration curve**.

22. The bacterial growth curve

Phase	What is happening
Lag	Cells adjusting — synthesising enzymes , taking up water and nutrients. Little or no cell division
Log (exponential)	Nutrients and space plentiful , waste low; cells divide by binary fission at the maximum rate ; population doubles at a constant interval
Stationary	Rate of division = rate of death ; caused by nutrients becoming limiting, toxic waste accumulating , lack of oxygen and space
	Death rate exceeds division rate ; nutrients exhausted, toxins

Death	accumulated
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The exponential growth rate constant — printed in the mark scheme

Mar 2024 Q7b(iii) gives the formula itself and four marking points:

$$k = \frac{\log N_t - \log N_0}{0.301 t}$$

• “indication of where **log values** have come from” — “e.g. $\log_{10}$ concentration/number read from a log graph, or calculated from actual numbers” • “during **log/exponential phase**” — “ACCEPT at end of lag phase and beginning of stationary phase” • “**determine t** / subtract the two time values” • “**growth rate equation used** to calculate the constant”

△ The values must be taken from the exponential phase. That is a mark in its own right.

Also usable: $N = N_0 \times 2^n$, so $n = \log(N/N_0) \div \log 2$, and generation time = $t \div n$.

Why a log scale is used: numbers span **several orders of magnitude**, and on a log scale the **exponential phase becomes a straight line** whose gradient gives the growth rate.

Growth in liquid culture — the practical: sterile **nutrient broth**, **cotton wool bung** (allows gas exchange, excludes microbes), inoculate aseptically, incubate at constant temperature, **shake or stir** to keep cells suspended and aerated, sample at regular intervals, measure by **turbidity — zero the colorimeter with uninoculated sterile broth**. Use **non-pathogenic** organisms, incubate at **≤25 °C**, **autoclave** before disposal.

When two students get different curves (June 2024 Q2): different **inoculum size** (shifts the start and lag), different **temperature** (changes the log-phase gradient), different **nutrient concentration** (changes the stationary population).

23. Bacteria and viruses

23.1 The comparison

	Bacteria	Viruses
Cell?	Prokaryotic cell	Not a cell

Genetic material	DNA , circular, plus plasmids	DNA or RNA
Protein coat	No	Capsid
Envelope	No	Some — from host membrane, with glycoproteins
Ribosomes	70S	None
Cell wall	Peptidoglycan	None
Reproduction	Binary fission , independently	Only inside a host cell

Two structures in all viruses: the **nucleic acid** and the **capsid**.

“Why must viruses be cultured in living cells?” (June 2019 Q3b(ii)) — the mark scheme:

- viruses cannot replicate **without a host cell** • they need the host cell to make **genetic material / for protein synthesis** • they need to use the **host cell’s enzymes, monomers and ribosomes** *“DO NOT ACCEPT to grow.”*

“Why are different viruses injected into different parts of the egg?” (June 2019 Q3b(iv)):

- each virus has a **specific host/target cell** • each virus has **attachment molecules/receptors** for the host cell — *“IGNORE antigens”* • host cells have **specific binding sites/receptors** that viruses bind to — *“IGNORE antigens”*

△ **“Antigens” is ignored throughout this answer.** Use **attachment molecules, receptors, binding sites**.

Why viruses are not in the three domains (Oct 2019 Q3b(i)): they are **not cells**, **cannot reproduce independently**, have **no metabolism** and **no ribosomes**, so they do not meet the criteria used for classification. The three domains are **Archaea, Bacteria and Eukarya**.

23.2 The four named viruses

Virus	Nucleic acid	Capsid	Envelope	Host
Ebola	RNA (negative sense)	Helical , filamentous	Yes	Humans, primates
TMV	RNA	Helical , rod	No	Plants

HIV	RNA + reverse transcriptase	Complex	Yes (gp120)	T helper cells
λ phage	DNA	Complex — head, tail, tail fibres	No	<i>E. coli</i>

The June 2019 Q3b(iii) mark scheme states flatly: “the host cell of HIV is the **T helper cell**.”

Ebola protein roles (Mar 2020 Q5) — the mark scheme: the **viral genome** codes for the viral proteins and is **replicated**; the **glycoproteins** allow **attachment to host cell receptors**; **VP24 and VP40** are involved in **virus assembly** and are **incorporated into the capsid**; **VP35** inhibits **interferon production**.

23.3 Lytic cycle and latency

Lytic: attachment → entry → **replication using host machinery** → assembly → **lysis/release**. **Latency:** the viral nucleic acid is **integrated into the host DNA** and **replicated with it**, producing **no new virus particles or symptoms**, until a trigger causes entry into the lytic cycle.

The application (Jan 2022 Q2c): during the symptomless period “new viruses are produced”, “viruses burst out of / damage host cells”, “infecting more cells / causing the spread of the virus” — the mark scheme adds “**IGNORE** refs to latency” in that particular question, so read whether latency is actually relevant.

RNA strand direction (Oct 2021 Q2b): a **negative-sense RNA strand cannot be translated directly**, so a **complementary positive strand must be made first**, which then acts as **mRNA** at the ribosomes.

24. TB and HIV

Tuberculosis — *Mycobacterium tuberculosis*

Transmission: droplet infection; aided by **overcrowding, poor ventilation, poor diet, immunosuppression**. **Infection:** inhaled into the lungs and **engulfed by macrophages by phagocytosis**. Its **thick waxy cell wall prevents the lysosome fusing with the phagosome**, so it **survives and reproduces inside the macrophage**. Infected macrophages are walled off in a **tubercle**, so the infection becomes **latent**, often for years. **Reactivation** when the immune system weakens; bacteria multiply and **destroy lung tissue**. **Symptoms:** persistent cough, coughing blood, chest pain, fever, night sweats, weight loss. **Death** from lung damage and reduced gas exchange, or from spread

to other organs.

HIV

Transmission: unprotected sex, contaminated blood/needles, mother to child (placenta, birth, breast milk). **Infection:** **gp120 glycoproteins bind to CD4 receptors on T helper cells**. **Reverse transcriptase** makes **DNA from the viral RNA**; **integrase** inserts it into the host genome, where it may be **latent**. On activation the host cell makes new viruses, which **bud out, destroying the T helper cell**. **Consequence:** T helper cell numbers fall. Since T helper cells **activate B cells and T killer cells**, the **whole specific immune response fails**. **AIDS** is diagnosed when the count falls below a threshold and **opportunistic infections** (TB, pneumonia) and rare cancers appear. **Death is from these secondary infections**.

The synergy (Jan 2024 Q5): HIV destroys T helper cells, so **latent TB reactivates**; TB activates the immune system, which **increases HIV replication**.

25. Barriers and non-specific responses

25.1 Barriers

Barrier	How it protects
Skin	Physical barrier of keratinised dead cells, continually shed; sebum lowers the pH
Stomach acid	pH ~2 denatures the enzymes and proteins of pathogens
Gut and skin flora	See below
Mucus and cilia	Mucus traps pathogens; cilia sweep it to the throat to be swallowed
Lysozyme in tears, saliva, sweat	Hydrolyses the peptidoglycan cell wall , so bacteria burst
Blood clotting	Seals wounds

Gut flora (Oct 2020 Q8b(i)) — the mark scheme's points:

- they **reduce/destroy/prevent the growth of pathogenic bacteria** — *"ACCEPT foreign/other bacteria/pathogens"*
- because they **compete for nutrients or space** —

"IGNORE food" • "ACCEPT produce vitamin K"

△ **"Compete for food" is ignored — say nutrients or space.**

Why antibiotics damage gut flora (Jan 2021 Q2c): many antibiotics are **broad-spectrum** and kill the **harmless gut bacteria as well as the pathogen**, because they share the **same target structures**. The flora then no longer outcompete pathogens, so species such as **C. difficile** multiply.

25.2 Inflammation

June 2019 Q2a — 2 marks, with two rejections:

• **increases blood flow / results in vasodilation** — *"ACCEPT histamine increases blood flow / permeability of capillaries"; "IGNORE capillaries/veins dilate"* • so **white blood cells / antibodies** are brought to, or leak out into, the site of infection — *"IGNORE macrophages if present in the blood"* • *"credit role of white blood cells in infection" — "e.g. antibodies for opsonisation, phagocytosis"*

△ **Say arterioles dilate or blood flow increases** — "capillaries dilate" is explicitly ignored.

The four cardinal signs: redness, heat, swelling and pain.

The full mechanism: mast cells release histamine → **vasodilation** of arterioles (redness, heat) and **increased capillary permeability**, so **plasma leaks into the tissues** (swelling, and pain from pressure on nerve endings) → more blood brings **more phagocytes and antibodies**; the raised temperature **inhibits pathogen reproduction**.

25.3 Interferon

Mar 2020 Q5d:

• **interferon is involved in the non-specific response against viruses** — *"ACCEPT inhibits viral replication"* • named mechanisms accepted: *"stimulates production of protein kinase, reduced protein synthesis, stimulates destruction of RNA, stimulates host cell (self) destruction, increases MHC I antigens on host cells"*

The application: Ebola's VP35 inhibits interferon production, so viral replication is not suppressed and the disease is more severe.

25.4 Phagocytosis

1. The phagocyte is attracted by **chemotaxis**.
2. It **recognises** the pathogen as non-self via **receptors** (aided by **opsonins**).
3. It **engulfs** the pathogen by **endocytosis** into a **phagosome**.

4. A **lysosome fuses**, forming a **phagolysosome**.
 5. **Hydrolytic enzymes digest** the pathogen — *"ACCEPT digestive enzymes / lysozyme"* (Aug 2022 mark scheme).
 6. The **macrophage presents the antigen** on its surface.
-

26. The specific immune response

26.1 The sequence — with the mark scheme's rejection

Aug 2022 Q3c(ii), 2 marks:

• *"macrophages present antigen (on surface / on MHC)"* • *"(macrophages present antigen) to T helper / CD4 cells"* — *"DO NOT ACCEPT to T killer cells / B cells / other cells"*

△ This is the single most-penalised error in Topic 6. Antigen presentation is to **T helper (CD4) cells**.

The full chain:

1. A **macrophage** engulfs the pathogen and **presents its antigen** on **MHC** — an **antigen-presenting cell**.
2. The antigen is presented to a **T helper cell** with a **complementary receptor**.
3. The T helper cell is **activated, divides by mitosis (clonal expansion)** and releases **cytokines**.
4. Cytokines **activate B cells** that have bound the same antigen, and **activate T killer cells**.
5. The B cell **divides by mitosis** into **plasma cells** and **B memory cells**.
6. **Plasma cells secrete antibodies** specific to that antigen.
7. **T killer cells** destroy **virus-infected or cancerous body cells** by releasing **perforin**.
8. **Memory cells persist**, giving long-term immunity.

26.2 The primary response delay — and the linkage requirement

Oct 2019 Q4b(i), 4 marks. The mark scheme carries an unusual instruction:

"NB answer must be linked with a time delay / many-staged process / slow process"

for max 4 marks to be awarded."

The creditable points:

1. this is the **first infection**, so no memory cells are present
2. **time taken for macrophages to bind, engulf and process** the virus
3. **time taken for presentation of antigen to T helper cells**
4. **time needed for T helper cells to activate B cells** — *"ACCEPT cause B cells to divide/proliferate"; "IGNORE activate T killer cells"*
5. **B cells need to differentiate into plasma cells**
6. **plasma cells need to synthesise antibody**

△ **Write it as a chain emphasising that each stage takes time.** A list of stages without the time framing is capped.

26.3 Primary versus secondary response

	Primary	Secondary
Delay	Long	Very short
Antibody concentration	Low	Much higher
Rate of production	Slow	Rapid
Duration	Short	Longer

Why the secondary response is faster and larger: memory cells specific to that antigen are already present in large numbers, so they are **activated immediately** and **divide rapidly into plasma cells**, destroying the pathogen **before symptoms develop**.

Why antibody concentration falls between exposures: antibodies are **proteins** and are **broken down**; plasma cells are **short-lived**.

△ **Specificity:** antibodies against virus P have **no effect** on virus Q, because their **variable regions are not complementary** to virus Q's antigens.

26.4 Antibodies

Structure: Y-shaped, **four polypeptide chains** (two heavy, two light), **disulfide bonds**, **two variable regions** forming the **antigen-binding sites**, a **constant region**, a **hinge region**.

How they work: agglutination (clumping pathogens, easier to engulf), **neutralisation** (blocking attachment, neutralising toxins), **opsonisation** (marking for phagocytosis), **activating complement**.

The rhesus/blood-group answer (June 2019 Q6b(ii)) — 4 marks:

• the **antigens are recognised as foreign** • and therefore initiate a **(humoral) immune response** • “credit details of humoral immune response” • resulting in **antibodies released by plasma cells** • “credit consequence” — “e.g. opsonisation / agglutination / destruction of RBCs (in liver/spleen/by phagocytes) / formation of memory cells”

27. Types of immunity

	Active	Passive
Natural	Infection	Antibodies from the mother — placenta, breast milk
Artificial	Vaccination	Injection of antibodies (antiserum)

The Oct 2019 Q4a table question — the mark scheme’s answers:

Feature	Answer
antigen-specific	both active and passive
provides long-term immunity	active only
antibodies injected	passive only

The full comparison:

Feature	Active	Passive
Antibodies made by	the individual’s own plasma cells	another organism
Memory cells	Yes	No
Speed	Slow	Immediate
Duration	Long-lasting	Short-lived

How a vaccine produces active immunity: the vaccine contains the **antigen** (dead,

attenuated or a fragment) which **does not cause disease**; it is **presented to T helper cells**, activating **B cells**, which divide into **plasma cells** (secreting antibodies) and **memory cells**; on **later exposure** the memory cells give a **rapid, large secondary response**.

📌 **Herd immunity (Jan 2021 Q3c) — 2 marks from three points:**

- **vaccinated people will not, or are less likely to, develop the disease** — ***"IGNORE less likely to be infected"***
- **fewer infected people to infect people who are not immune** — ***"ACCEPT herd immunity"***
- **protects people who cannot become immune** — ***"e.g. allergic to vaccines, immunodeficient, immunosuppressed"***

Why people refuse vaccination: fear of side effects; religious or cultural objection; distrust of governments or companies; belief the disease is rare or mild; misinformation; cost or access.

28. Pathogen evasion — the evolutionary race

Hosts evolve better defences and pathogens evolve better evasion; each is a **selection pressure** on the other. Pathogens usually evolve faster because of their **short generation time and high mutation rate**.

Mechanism	Example
Antigenic variability	Influenza and HIV change surface antigens, so memory cells no longer recognise them
Hiding inside host cells	TB inside macrophages ; viruses inside cells where antibodies cannot reach
Preventing lysosome fusion	TB's waxy cell wall
Latency	Herpes, HIV integrate into host DNA
Destroying immune cells	HIV destroys T helper cells
Suppressing the response	Ebola VP35 inhibits interferon
Capsules	Prevent recognition and phagocytosis

29. Antibiotics

🔑 The definition (June 2019 Q2b(i)) — 2 marks with two rejections:

• “**substance / chemical / drug / medicine**” — “**DO NOT ACCEPT produced by a virus**” • “**that kills or inhibits the growth of bacteria / microorganisms / prokaryotes**” — “**ACCEPT bacteriostatic/bactericidal**”; “**DO NOT ACCEPT virus**”; “**IGNORE pathogens**”

⚠ Three separate traps in one 2-mark question. Antibiotics are produced by **microorganisms** (not viruses), act on **bacteria** (not viruses), and the word “**pathogens**” is ignored.

“Why do we need alternatives?” (June 2019 Q2b(ii)):

• because **bacteria are becoming resistant** — “**DO NOT ACCEPT immune**”

	Bacteriostatic	Bactericidal
Effect	Inhibits growth and reproduction	Kills
On a graph	number stays constant (plateau)	number falls
Examples	tetracycline, erythromycin	penicillin, vancomycin

How antibiotics work, and why not on viruses or human cells:

They target structures **unique to bacteria** — the **peptidoglycan cell wall** (penicillin prevents cross-linking, so the cell bursts by osmosis), **70S ribosomes**, **DNA gyrase**, **folate synthesis**. Human cells have **no cell wall** and have **80S ribosomes**. **Viruses have no cell wall, no ribosomes and no metabolism of their own**, so there is **nothing to target**.

The antibiotic-effect practical (mast ring / disc diffusion)

1. Spread a **lawn** of bacteria on sterile agar using aseptic technique.
2. Place **antibiotic discs** or a **mast ring, evenly spaced**, with **sterile forceps**.
3. Include a **control disc** of **sterile water or solvent only**.
4. **Tape (not seal)**, **invert**, incubate at **25 °C for 24–48 h**.
5. Measure the **diameter of the clear zone in two directions at right angles**; take a mean; convert to **area** using πr^2 .

A larger clear zone = a more effective antibiotic, because it **diffused further while still at an inhibitory concentration**. **Control variables**: same bacterial species and concentration, same agar, same antibiotic concentration per disc, same temperature and time. **Limitations**: antibiotics **diffuse through agar at different rates** (larger molecules more slowly), so zone size is not potency alone; results *in vitro* may not predict effectiveness **in the body**; you **cannot tell bacteriostatic from bactericidal** from the zone alone without subculturing.

Hospital-acquired infections (MRSA, *C. difficile*): *Causes*: many **ill and immunocompromised** people together; **invasive procedures** bypassing the skin; **overuse and misuse of antibiotics**; courses **not completed**; antibiotics prescribed for **viral** infections; poor **hand hygiene**; staff moving between patients. *Codes of practice*: prescribe **only when necessary** and **not for viral infections**; use **narrow-spectrum** antibiotics; ensure the **full course** is completed; **rotate** antibiotics; **hand washing** before and after every patient; **isolate** infected patients; **screen** on admission; **sterilise** equipment; **educate** staff and visitors.

30. Decomposition and carbon recycling

Saprotrophic (extracellular) digestion — the chain:

1. The decomposer **secretes enzymes** onto the dead material, packaged by the **Golgi apparatus** into vesicles and released by **exocytosis**.
2. Enzymes catalyse the **hydrolysis** of large insoluble polymers — **carbohydrases** → **sugars**, **proteases** → **amino acids**, **lipases** → **fatty acids and glycerol** — **outside the organism**.
3. The **small soluble products are absorbed**.
4. The decomposer **respires** them, **releasing carbon dioxide to the atmosphere**.

The synoptic organelle question (June 2019 Q1c–d): the **Golgi apparatus modifies and packages the digestive enzymes into vesicles** for secretion; the **mitochondria** carry out **aerobic respiration**, producing the **ATP** needed for enzyme synthesis and active transport, and **releasing CO₂** which returns carbon to the atmosphere.

Why fungi are well adapted: **hyphae/mycelium** give a very **large surface area** for secreting enzymes and absorbing products, and can **penetrate** the material.

Detritivores (earthworms, woodlice) **break material into smaller pieces**, increasing the **surface area** for decomposers.

Rate of decomposition depends on **temperature, oxygen availability** (aerobic → CO₂, faster; **anaerobic** → **methane**), water, pH and the material.

Why peat bogs preserve material: waterlogged (anaerobic) and acidic, so decomposer activity is very slow — which is why **pollen is preserved** (§11).

31. PCR

The three steps:

Step	Temperature	What happens
Denaturation	95 °C	Hydrogen bonds break , separating the two strands
Annealing	50–65 °C	Primers bind to the ends of the target sequence by complementary base pairing
Extension	72 °C	Taq DNA polymerase adds free nucleotides by complementary base pairing

The reagents (Mar 2022 Q5b(i)) — any two, with three explicit rejections:

• (DNA) primers • (DNA) mononucleotides — “ACCEPT all 4 listed”; “**IGNORE bases**”; “**DO NOT ACCEPT RNA nucleotides**” • Taq / DNA polymerase — “ACCEPT taq enzyme”; “**DO NOT ACCEPT RNA polymerase**” • buffer

⚠ **Three ways to lose this mark:** writing “bases”, “RNA nucleotides” or “RNA polymerase”.

Why Taq: it is from a **thermophilic bacterium**, so it is **not denatured at 95 °C** and need not be replaced each cycle.

`copies after n cycles = 2n × starting number`

Timing calculation (Mar 2022 Q5b(ii)): multiply the cycle time by the number of cycles — the mark scheme’s answer, “4950 seconds / 82.5 minutes / 1.375 hours”, shows any consistent unit is accepted.

Value: allows analysis from **tiny samples** — a hair, a drop of blood, **environmental DNA (eDNA)**. **Risk:** it amplifies **any contaminating DNA** just as powerfully, giving false results.

32. Gel electrophoresis and DNA profiling

The method, and why it works:

1. DNA is cut with **restriction endonucleases**, which cut at **specific base sequences**, producing **fragments of different lengths**.
2. Samples are loaded into **wells** in an **agarose gel** covered in **buffer**.
3. A **voltage is applied**. DNA is **negatively charged because of its phosphate groups**, so all fragments move **towards the positive electrode (anode)**.
4. **Shorter fragments move faster and further**, because they meet **less resistance** and pass more easily through the pores in the gel.
5. Bands are visualised by **staining** or by **radioactive/fluorescent probes and autoradiography**, and sized against a **DNA ladder**.

Gel concentration (Jan 2021 Q7c): a **higher agarose concentration** gives **smaller pores**, so fragments move **more slowly** and short fragments are better separated.

Circular vs linear DNA (Jan 2021 Q7d): **circular DNA moves faster** because it is **more compact/supercoiled**. Examples in cells: **plasmids, bacterial chromosomal DNA, mitochondrial DNA, chloroplast DNA**. Differences: circular has **no free ends, no histones**, and is generally **shorter**.

DNA profiling — the principle: most of the genome is **non-coding** and contains **short repeated sequences** whose **number of repeats varies between individuals**, so the band pattern is effectively unique.

EXTRACT DNA → AMPLIFY by PCR → CUT with restriction enzymes → SEPARATE by electrophoresis → VISUALISE → COMPARE band patterns

Interpreting profiles:

- **Forensics:** all bands matching = a match.
- **Paternity:** every band in the child must be present in **one parent or the other**.
- **Species identification and relatedness:** the **more bands in common, the more closely related** — used for the peat-bog pollen (Oct 2019 Q2b(ii)), lemur origins (June 2021 Q4b) and illegal wildlife trade (Jan 2026 Q4).

Why non-coding DNA is used: it is **highly variable between individuals**, because there is **no selection pressure** keeping it constant.

33. Determining time of death

The mark schemes are unusually explicit about the **limitations**, and those are where the marks are.

33.1 Body temperature (algor mortis)

After death **metabolism stops**, so **no heat is generated** and the body **cools towards ambient temperature** — roughly **1.5–2 °C per hour** at first, following a **sigmoid** curve. Measure a **core** temperature (rectal, ear, under the liver).

Effect of a cooler environment (Jan 2021 Q2b(ii)) — 2 marks:

- the estimate would be **shorter / an under-estimate** • because a **body loses heat faster in cooler conditions**

Factors affecting cooling rate: **ambient temperature; body size and surface area to volume ratio; body fat** (an insulator); **clothing/covering; wind, humidity, immersion in water;** the person's **temperature at death**.

Limitation: useless once the body has **reached ambient temperature** — the Oct 2019 Q5c mark scheme says *“measuring body temperature is not accurate as it would have levelled off / reached ambient temperature.”*

33.2 Rigor mortis

After death **respiration stops**, so **no ATP is produced**. Without ATP the **actin–myosin cross-bridges cannot detach**, so muscles remain contracted and **stiff**. Rigor begins in the **smallest muscles first** (eyelids, face, neck) after about **4–6 hours**, is **complete at ~12 hours**, and **passes off after ~24–36 hours** as the body's own **enzymes break down the muscle proteins**.

“Why is body stiffness alone insufficient?” (Jan 2021 Q2c) — 3 marks, the mark scheme's exact points:

- because **temperature affects rigor/body stiffness** — *“ACCEPT exercise / body shape / body fat / ATP levels”* • because **deciding when a body is stiff or not stiff is subjective** — *“ACCEPT gives a wide range of time values”* • because **if the body is stiff, the time since death can only be estimated as being between 3 and 36 hours** — *“ACCEPT if not stiff cannot tell how many hours after 36 hours”*

And from Oct 2019 Q5c: *“the muscles become stiff and then relax again”* — *“ACCEPT rigor mortis does not last long.”*

⚠ **The ambiguity point is the key mark:** the same degree of stiffness occurs **twice** —

during onset and during release.

33.3 Extent of decomposition

Autolysis (the body's own enzymes) then **putrefaction** by bacteria, producing gases, bloating, discolouration, skin slippage and finally skeletonisation.

Its limitations (Oct 2019 Q5c):

- it depends on ambient temperature, presence of oxygen, presence of water and location
- it depends on the types and numbers of decomposers that colonise the body

33.4 Forensic entomology

Insects colonise the body in a **predictable sequence**. **Blowflies** are attracted within minutes to hours and lay eggs in **moist areas — eyes, nose, mouth, wounds**. Eggs hatch into **larvae** which pass through **instars** of known size, then **pupate**. Each stage takes a **known time at a known temperature**. By **identifying the species and stage — or measuring larval length — and using temperature records**, the **minimum time since death** is calculated.

Why temperature data are essential (Oct 2019 Q5a): larval development is **enzyme-controlled**, so it is **faster at higher temperatures**; the same larval length means a different age at a different temperature.

Limitations from the mark scheme (Oct 2019 Q5b(ii)):

- “do not know how far into the prepupal stage” the larvae are
- other factors besides ambient temperature affect development — “e.g. body temperature, pH, water availability, oxygen availability”
- the body may have been **moved**, so local records may not apply
- **insects may have been excluded** (indoors, buried, wrapped, submerged), giving an **underestimate**

⚠ **Forensic entomology is NOT “the most accurate method”** (Oct 2019 Q5c asks you to evaluate this claim). It gives only a **minimum** time since death and depends on insect access and temperature assumptions.

33.5 Stage of succession — with the named species

Mar 2022 Q8b(v), 3 marks — the mark scheme names the sequence:

- **succession means that the species of insects found on the decomposing mammal will change with time / appear in a specific sequence**
- ***Lucilia* appears on the body first — “ACCEPT pioneer species”**
- **then *Cochliomyia* / *Chrysomya* appear and *Lucilia***

numbers decrease / are outcompeted • then *Ophyra* arrives and
Cochliomyia/Chrysomya decrease / are outcompeted

Why eggs are collected and identified (same question): “to identify the species that laid the eggs” and “to determine time until hatching, to find out when they were laid.”

33.6 The evaluation answer

Any **single method gives only a range**, not a precise time; the assumptions (37 °C at death, constant ambient temperature, immediate insect access) may not hold; the body may have been **moved**. Therefore **several methods should be used together**, each accurate over a **different time window**, and the estimate given as a **range**.

PART D — EXAM TECHNIQUE

34. Levels-based marking on * questions

Mark schemes give **indicative content** plus a levels grid:

Level	Marks	Descriptor
0	0	No awardable content
1	1–2	<i>“Demonstrates isolated elements of biological knowledge and understanding to the given context with generalised comments made. Vague statements.”</i>
2	3–4	Some accurate knowledge applied to the context; some logical linkages ; reasoning partially developed
3	5–6	Accurate and detailed knowledge applied throughout; coherent, logically structured , with sustained linkages

And where a question has data, the mark scheme tags each point (D) or (E) and specifies the ratio. The sand-dune scheme (§10) is explicit:

“5 marks = 4 descriptions of the data and all 4 explained; 6 marks = 5 descriptions and all five explained.”

So the winning structure for a data-based * question is a paired list:

DESCRIBE the trend (with a figure) → EXPLAIN why it occurs →
 DESCRIBE the next trend → EXPLAIN why → ...

For a non-data * question, write **one linked argument**, applying the biology to the **specific context** — “in this ecosystem”, “for these bacteria”, “because these dunes are older”. Connectives are the mechanism: **because · so · this means that · therefore · as a result · which leads to · eventually**.

⚠ **Six correct but unlinked facts score 4, not 6.**

35. Command words

Command	What the mark scheme wants
State / Give / Name	A short fact. No explanation
Describe	What happens or what the data show , with figures quoted . No reasons
Explain	Reasons — must contain <i>because / so / due to</i>
Suggest	Apply biology to an unfamiliar context; more than one answer usually accepted
Calculate	Working, unit , correct significant figures , correct rounding
Deduce	A conclusion from the data , with the reasoning
Compare / Contrast	Comparative language for both — “X is higher than Y”
Comment on	A judgement supported by evidence — usually reliability or validity
Discuss / Evaluate	Both sides , then a conclusion
Identify conclusions that can be drawn	Statements supported by the data ; do not go beyond it , and do not explain

“Describe the relationship shown in the graph” has a fixed formula:

State the **overall trend**; quote **figures with units**; say whether it is **linear**; identify any **plateau, peak or change of direction with the value at which it occurs**; mention any

36. Calculations that recur

% change	= (change ÷ original) × 100	– state INCREASE or DECREASE
% of a total	= (part ÷ total) × 100	
efficiency	= (energy in ÷ energy in previous level) × 100	
rate	= change ÷ time, with a unit	
Q_{10}	= rate at (T+10) ÷ rate at T	– NO UNITS
k	= (log N_t – log N_0) ÷ (0.301 t)	– from the LOG PHASE
magnification	= image ÷ actual	– convert mm to μm

Worked from the mark schemes themselves:

- **Percentage of a group (Oct 2020 Q9b(i)):** *“total number of moose added up and total with 50 000 or more ticks calculated (214 and 41)”, then “percentage calculated to max 2 dp: $41 \times 100 \div 214 = 19 / 19.16 / 19.2$ ” — and note “CE applies if only one of the two numbers is incorrect”, so a single arithmetic slip still allows the method mark.*
- **Efficiency (June 2019 Q5a(iii)):** $(6 \times 100) \div 67 = 8.96$.
- **Ratio (Oct 2019 Q6a):** $1 : 9.5$ — “ACCEPT $1 : 9.54 / 10 : 95 / 0.1 : 1$; IGNORE correct fractions.”
- **pH is a log scale (June 2021 Q7a(i)):** *“a difference of 2 is 100-fold”, and “there are fewer hydrogen ions at pH 8.5 than pH 6.5.”*
- **Percentage increase (June 2021 Q7a(ii)):** answer **650 %** — remember that a percentage *increase* can exceed 100.

Significant figures and rounding are marked. Follow the precision of the data, and round correctly.

37. Mark-scheme phrase bank

Photosynthesis

- *“to absorb light so that the **electrons will be excited / released from photosystems**”*
- *“so that **reduced NADP and ATP** can be synthesised”*
- *“**water is split by light** / reference to **photolysis**”*

- *"RuBP regenerated too slowly to bind as much CO₂"*
- *"amino acids made from **glucose and nitrates**"*
- **C and O from carbon dioxide; H from water** — *"DO NOT ACCEPT O from water"*

Ecology and climate

- *"organisms interacting with each other **and with the abiotic environment**"*
- *"niche is the **role that an organism plays**" + a stated reason*
- *"(anthropogenic) **caused by the effects of humans**" + "changes to mean temperature or rainfall"*
- *"**anaerobic conditions result in methane** being produced"*
- *"methane **traps infra-red radiation**" → "increase in temperature of the atmosphere/surface/water"*
- *"the **trend could change**" | "the data **fluctuates**"*

Evolution

- *"**mutation** causing an **allele** for resistance"*
- *"the chemical acts as a **selection pressure**"*
- *"resistant individuals **survive and pass the allele to their offspring**"*
- *"**reduced gene flow** → changes in **allele frequency** → **reproductively isolated** → two species"*

Microbiology and immunity

- *"on agar / in a broth" + "aseptic technique" + "incubated at **25–35 °C**"*
- *"they **compete for nutrients or space**"*
- *"**increases blood flow / vasodilation**" → "white blood cells and antibodies brought to the site"*
- *"macrophages **present antigen ... to T helper / CD4 cells**"*
- *"**B cells differentiate into plasma cells**" → "plasma cells **synthesise antibody**"*
- *"viruses need the **host cell's enzymes, monomers and ribosomes**"*
- *"substance that **kills or inhibits the growth of bacteria**"*

Forensics

- *"the body **loses heat faster in cooler conditions**"*
 - *"deciding when a body is stiff is subjective"*
 - *"**succession** — the insect species **appear in a specific sequence**"*
-

38. The twenty most commonly lost marks — each one a mark-scheme rejection

1. **"Immune" instead of "resistant"** — *DO NOT ACCEPT immune for resistant.*
2. **"The selection pressure caused the mutation"** — *DO NOT ACCEPT.*
3. **Presenting antigen to T killer or B cells** — it is **T helper (CD4)** cells; *DO NOT ACCEPT* the others.
4. **"Grown in a Petri dish"** — *IGNORE*; name **agar or broth** and state a **temperature**.
5. **Saying antibiotics are produced by a virus, or act on viruses** — both *DO NOT ACCEPT.*
6. **"Pathogens"** in the antibiotic definition — *IGNORE*; say **bacteria**.
7. **"Capillaries dilate"** in inflammation — *IGNORE*; say **blood flow increases / vasodilation**.
8. **"Compete for food"** for gut flora — *IGNORE*; say **nutrients or space**.
9. **"O from water" in glucose** — *DO NOT ACCEPT*; oxygen in glucose comes from **CO₂**.
10. **Listing the primary-response stages without the time framing** — the scheme caps this.
11. **Defining ecosystem without the abiotic environment** — *IGNORE habitat.*
12. **Defining niche without a stated reason** when the question gives a context.
13. **Naming human activities** for "anthropogenic" — *IGNORE named activities.*
14. **"Global temperatures"** instead of the **atmosphere/surface/water** — *IGNORE.*
15. **Q₁₀ given with units** — *DO NOT ACCEPT with units.*
16. **"Bases" or "RNA nucleotides" or "RNA polymerase"** in PCR reagents — all rejected.
17. **"To grow"** as the reason viruses need host cells — *DO NOT ACCEPT*; say they need the host's **enzymes, monomers and ribosomes**.

18. **"Antigens"** where the question means **attachment molecules / receptors** — *IGNORE*.
19. **Describing data without explaining it** in a levels-based question — caps you at Level 1–2.
20. **Going beyond the data** when asked for conclusions that can be drawn.

39. Per-paper map

Paper	Question topics
2019 Jun	Decomposition (Golgi, mitochondria) · sepsis, inflammation, antibiotics, nanowires · culturing bacteria and viruses · whale sharks and eDNA* · GPP/NPP and efficiency; starch vs cellulose · rhesus antigen* · photolysis, GALP, net CO ₂ uptake · sea ice, extrapolation, polar bears
2019 Oct	GALP → lipids · pollen in peat bogs, DNA profiling · three domains, giant viruses, tRNA · four types of immunity, antibody graphs · forensic entomology and its limits · wolves and moose* · giraffe weevil: endemism, evolution, speciation* · cattail, methane, anaerobic decomposition
2020 Mar	NPP and ecosystem · plastics and microbial evolution · temperature records, tree rings · rainforest loss, mimicry · Ebola: genome, glycoproteins, VP24/40, VP35 and interferon · barnacles and niche; reliability · seaweed pigments, GALP → protein, sustainability* · T helper/T killer, CAR-T · turbidity and calibration curves*
2020 Oct	Chloroplast, absorption/action spectra, Rf · body temperature and stiffness · photolysis, bond energies, amino acids · rheumatoid arthritis, TNF · bacteriophages, holins, lysins · squirrel investigation design* · fungi vs bacteria, antibiotic prescriptions · blood groups, gut flora · moose, ticks, global warming*
2021 Jan	Inflammation, curcumin · antibiotics and gut flora · vaccination, herd immunity · greenhouse gases, distribution, beetles · ATP, LDR, LIR · zebra stripes, fly resistance* · gel electrophoresis, circular DNA · quince browning, Q ₁₀ , decomposition
2021 Jun	Moss species and genome size · chloroplast vs mitochondrion; cyclic photophosphorylation · habitat/population/community/niche; efficiency · lemurs, PCR, allopatric and sympatric speciation · larval length, body temperature · carbon cycle, anthropogenic change · sand dune succession*; NPP · cystic fibrosis, phage therapy*
2021 Oct	Chloroplast and light/dark · rabies: envelope, capsid, RNA strands · four types of immunity, macrophages · ecology and forensics

2022 Jan	Polar bear · thylakoid membranes and ATP synthase · HIV · pathogen evasion · crow speciation · chloroplast structures · food demand · forensic entomology
2022 May	Bacteriostatic vs bactericidal · growth curve · skin flora · regent honeyeater · trophic energy · HIV structure · Tasmanian devil · volcanic succession
2022 Oct	Growth curve · virus classification · chloroplast, LDR/LIR · mud cores · globular protein · Alzheimer's · MRSA · forest fires and biodiversity* · biomass above and below ground
2023 Mar	Mast ring resistance · photosynthetic bacteria · vanilla · natural rubber · global warming · temperature and enzymes · DNA and endangered species · dengue · TB
2023 Jun	Magnesium uptake · black bear · tetracycline vs vancomycin · giant panda · PBF · malaria · whaling · reforestation · time of death
2023 Oct	Chloroplast · non-specific and specific responses · growth in liquid culture · breast milk · seaweed · telomeres · cancer immunotherapy · trophic levels · sea turtles
2024 Jan	Skin flora · light-independent reactions · apple maggot sympatric speciation · glaciers · TB and HIV · rainforest food chain · yeast · artificial immunity
2024 Mar	Q_{10} and C_3/C_4 · speciation · growth rate constant · (plus standard topics)
2024 May	Seaweed pigments · two students' cultures · textiles · leaf litter · herpes and latency · aquaculture · CO_2 and photosynthesis · Bajau divers
2024 Oct	Light absorbed by a field · broccoli · T killer cells · time of death · succession · RSV · leaf scald · phytoplankton · light and photosynthesis
2025 Jan	Photosynthesis–productivity · time of death · lynx and hare · chloroplast and Calvin cycle · peat bogs · antibiotic resistance · greenhouse gases · cholera · endogenous retroviruses
2025 May	Redonda succession · antibiotics and gut flora · body temperature · tigers · antibodies · chloroplast DNA · beta-carotene · reptile eggs and temperature · rice yellow mottle virus
2025 Oct	Pathogens · LDR and LIR · coral reefs · trophic energy · cyanobacteria · antibiotic resistance · nematodes · infectious disease
2026 Jan	Energy transfer · temperature and photosynthesis · sea ice · forensics and illegal wildlife trade · kerosene and biofuels · porcine circovirus · dolphins · microplastics
2026	HIV and TMV · Kaibab squirrel speciation · <i>Fusarium</i> · chloroplast pigments · plant

The architecture is fixed. Every paper has a **photosynthesis** question, an **immunity** question, a **microbial growth or antibiotics** question, a **climate change** question and an **energy flow** question, plus usually **evolution/speciation** and **forensics**. Prepare those seven and you have 70–80 of the 90 marks.

40. What an A* requires, and the final plan

The rule

A* requires **grade A for the IAL overall** (≥ 480 of **600** uniform marks) **and** $\geq 90\%$ of the uniform marks across the **IA2 units combined** (≥ 270 of **300**).

The IA2 units are **4, 5 and 6** — $120 + 120 + 60$ UMS. **The A* is decided entirely by A2 performance.** Unit 6 is **60 of the 300**, and it is examined on the practical work behind Units 4 and 5 — so §6, §9, §16, §20–22 and §29 of this note are Unit 4 content *and* Unit 6 content.

Priority order

Priority	Content	Marks/paper
1	Photosynthesis chains: LDR, Calvin cycle, GALP products (§3, §4)	12–20
2	The immune response sequence and the primary-response delay (§26)	10–18
3	The definitions with their rejections (§8, §20, §27, §29)	8–12
4	Growth curve, four phases, growth rate constant (§22)	6–12
5	Evolution and speciation chains, with the two rejections (§17, §18)	5–12
6	Climate evidence and the greenhouse chain (§11, §12)	8–16
7	Time of death and its limitations (§33)	4–10
8	Levels-based * technique — pair (D) with (E) (§34)	10–15
9	Practical method marks (§6, §9, §20, §29)	6–12

Five-step routine for every question

1. Circle the COMMAND WORD and the MARK ALLOCATION — how many statements do you o
2. Underline the CONTEXT and any data — Level 3 answers use them
3. Calculations: working → answer → UNIT → significant figures → correct rounding
4. Explanations: write the required number of statements, LINKED, using \$37's phr
5. Re-read: has it asked for a reason, a comparison, a conclusion or a LIMITATION that you have not given?

Exam-day tactics

- **Pace by the mark allocation** — about a minute a mark plus a little.
- **Bring the HB pencil** for graphs and diagram completion.
- On * questions with data, **alternate description and explanation**.
- **Quote figures with units** whenever describing data.
- Never leave a **"suggest"** question blank — it is designed to be answerable from general biology.
- Leave five minutes to check that every *explain* answer contains a **because**, every calculation has a **unit**, and every *evaluate* has a **conclusion**.